

and Why?

Overview Research consistently shows that the best approach for lower-back pain is alternating between sitting and standing throughout the day, rather than doing only one or the other. Prolonged static postures - whether sitting or standing - both present risks for musculoskeletal discomfort and lower-back pain (LBP).

Key Findings from Recent Research A Griffith University-led study (to be published in *Applied Ergonomics*, February 2026) found that a structured schedule of 30 minutes sitting followed by 15 minutes standing reduced lower-back pain more effectively than personalized, ad-hoc sit-stand schedules in desk workers. A separate systematic review and meta-analysis indicates that prolonged standing is not necessarily superior to sitting for back pain relief. However, the SUFHA trial and a 2024 review of office workers found reduced musculoskeletal discomfort and lower-back pain scores when sit-stand desks were used compared with control groups.

Standing-Only Desk Standing all day can increase lower-back compression and leg fatigue if done continuously. Prolonged standing for desk work does not result in lower ratings of perceived low back pain than sitting, according to systematic review evidence. Standing desks alone are not a cure for back pain.

Sitting-Only Desk Sitting all day promotes static loading on spinal discs and weakens supporting muscles over time. Extended sitting without movement breaks is associated with increased musculoskeletal discomfort and poorer health outcomes.

The Evidence-Based Solution: Alternating Postures The optimal strategy supported by current research is using a sit-stand desk with a structured alternating schedule. The 30:15 ratio (30 minutes sitting followed by 15 minutes standing) appears to be the evidence-backed sweet spot for reducing lower-back pain while maintaining productivity.

Additional Recommendations Movement breaks (walking, stretching) every 30-60 minutes remain valuable regardless of desk setup. Posture quality in both positions matters more than the desk type itself. Individual optimal ratios may vary slightly based on personal ergonomics and health status.

Conclusion Rather than choosing solely a standing or sitting desk, invest in a sit-stand desk that allows seamless alternation throughout the workday. Follow a consistent schedule such as 30:15 sitting-to-standing, take regular movement breaks, and maintain good posture in both positions to minimize lower-back pain risk.

References 1. Does fixed or personalised sit-stand desk ratios improve lower back pain? *Applied Ergonomics*, 2026. [sciencedirect.com] 2. Low back pain and sitting time, posture and behavior in office workers. *Human Factors*, 2025. [journals.sagepub.com]